Supplements: what works and what is a waste of money

The shelves are packed and the marketing promises miracles. The reality is duller: a few supplements have solid evidence behind them, and most of the rest just empty your wallet. Here is what is worth paying for, and what to leave on the shelf. Practical, and based on the research.

Basics first, pills second

Before you think about any supplement, get the order straight. No supplement makes up for a poor diet, too little sleep and random training. A supplement is **the cherry on top**, not the cake. The professional bodies are clear on this: the vast majority of the effect comes from total energy intake, protein and training. Supplements add a few percent on top, at best [1,6].

> **Main point:** Supplement = an add-on. If calories, protein and sleep are off, no expensive powder will save it [1]. Get food and recovery in place first. Only then spend money on supplements.

What actually works

This is the short list of supplements with a repeatedly confirmed effect, the ones worth your money:

- ✓ **Creatine monohydrate (3–5 g/day):** the best studied and most effective legal supplement for strength and muscle mass. Take it daily. A steady dose is enough. A "loading phase" is not required [1,2]. Cheap, and safe with long-term use [2].
- ✓ **Protein / whey:** a practical way to fill the day's protein. Adding protein increases the strength and muscle you get from training [5]. The target of about 1.6–2.2 g/kg per day can come from food. The powder is only convenience [4,5].
- ✓ **Caffeine (3–6 mg/kg, about 30–60 min before the session):** reliably raises performance, strength and endurance, and lowers perceived effort [3]. It works as a capsule, a tablet, or a strong coffee [3].
- ✓ **Vitamin D (the dose depends on your blood level):** it makes sense mainly when you are low. A low level is common in winter if you get little sun. Supplementing lowers the risk of acute respiratory infections, most in people whose level is low [7]. Ideally get the level measured and dose from that, not by guessing. This guide does not give you a dose to take blind.

- ✓ **Omega-3 (EPA+DHA):** if you don't eat fish 2 times a week, EPA+DHA has reasonable support for health and for the inflammatory profile. Look at the EPA+DHA on the serving, not just "fish oil" in total.
- ✓ **Beta-alanine (about 3-6 g/day):** helps on repeated hard efforts lasting about 1-4 minutes [8]. For low-rep strength work it is a side note. The skin tingling is harmless [8].

What is a waste of money

Now the other side. These categories usually promise more than they can deliver, and the money is almost always wasted:

- ✓ **"Fat burners":** an expensive mix of caffeine and herbs with a minimal real effect on body composition. Fat is burned by a calorie deficit, not by a capsule [1]. The part that works is caffeine. You already have that in coffee, and coffee is cheap [3].
- ✓ **BCAAs when protein is already high enough:** if you hit your daily protein, extra BCAAs add nothing. A complete protein already contains all the amino acids [4]. Wasted money for almost anyone who eats enough protein.
- ✓ **"Detox" teas and cleanses:** your liver and kidneys do the detoxing, and no tea helps them do it. A marketing word with no support.
- ✓ **Testosterone boosters:** herb blends that promise a "natural testosterone rise" do not, in practice, work in healthy people in a way that means more muscle or more strength. Muscle is built by training, food and sleep.
- ✓ **Pre-workouts with a "proprietary blend":** often just expensive caffeine hiding behind a secret mix. If you want the effect, use caffeine on its own, at a clear dose [3].

> **Rule of thumb:** If a product promises a fast, dramatic result with no work, it is almost certainly marketing. The supplements that work are boring, cheap and tested for years [1,6].

Watch purity and contamination

Supplements are not regulated like medicines. That means what is in the tub may not match the label. And a share of products on the market are contaminated with substances that have no business being there. The consensus of sports medicine warns that contamination, or undeclared substances in supplements, is a real risk, including substances on the doping list [6].

- ✓ Buy from **established brands** with independent testing (for example third-party certification) [6].

- ✓ The **longer and more secret the formula** (exotic names, a "proprietary blend"), the more careful you should be.
- ✓ If you are a tested athlete, use only products certified against banned substances. The risk of unintentional doping is real [6].
- ✓ Stick to **simple, single-ingredient** products (plain creatine monohydrate, whey, caffeine). Less room for a surprise.

The doses, so you don't have to look them up

A short cheat sheet for the things that make sense. Stay inside the ranges that have been tested. More is not better. Nothing here is a medicine dose, and nothing here is personal medical advice.

1 Creatine
3-5 g a day, every day, any time of day. No loading phase. Cheap, safe, well tested [2].

2 Caffeine
3-6 mg per kg of body weight, about 30-60 min before the session [3]. Late afternoon, watch your sleep.

3 Vitamin D
The dose follows a measured blood level, not a guess. It makes sense mainly if you are low, typically in winter [7].

4 Omega-3
Look at EPA+DHA per serving, not total "fish oil". It makes sense mainly if you don't eat fish.

5 Protein
Use it to fill a daily target of about 1.6-2.2 g/kg [4,5]. The powder is convenience, not a duty. Food can do it.

6 Beta-alanine
About 3-6 g a day. The effect is on efforts of 1-4 min [8]. Split the dose so the skin tingles less.

Order of priority (food and sleep first)

Before you spend anything on a supplement, walk this list from the top. Most people only need the first few points, and they start caring about supplements too early:

- ✓ **1. Energy and protein for the day.** Hitting your calories and about 1.6-2.2 g/kg of protein does more than a whole cabinet of supplements [4,5].
- ✓ **2. Sleep and recovery.** The cheapest "supplement" for performance and for fat loss. Without it you sabotage even a perfect diet. See Sleep and recovery.
- ✓ **3. Consistent training.** No powder replaces regular, progressive loading.
- ✓ **4. Only now, the supplements with evidence.** Creatine, protein, caffeine, vitamin D according to your blood level, omega-3 [1,2,3,7].

✓ **5. Leave the marketing.** Fat burners, extra BCAAs, detox, testosterone boosters: put that money into decent food instead [1].

Action step: This week, don't buy anything new. First check whether you are hitting **calories, protein and sleep**. If you are, and you want to add something, take **creatine (3-5 g/day)** [2] and, if you like, caffeine before training [3]. The rest is mostly marketing. Be Effective! 💪

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Sources: [1] Kerksick CM, et al. ISSN exercise & sports nutrition review update: research & recommendations. J Int Soc Sports Nutr. 2018;15:38. [2] Kreider RB, et al. ISSN position stand: safety and efficacy of creatine supplementation. J Int Soc Sports Nutr. 2017;14:18. [3] Guest NS, et al. ISSN position stand: caffeine and exercise performance. J Int Soc Sports Nutr. 2021;18:1. [4] Jäger R, et al. ISSN position stand: protein and exercise. J Int Soc Sports Nutr. 2017;14:20. [5] Morton RW, et al. Protein supplementation and resistance training-induced gains: meta-analysis. Br J Sports Med. 2018;52(6):376-384. [6] Maughan RJ, et al. IOC consensus statement: dietary supplements and the high-performance athlete. Br J Sports Med. 2018;52(7):439-455. [7] Martineau AR, et al. Vitamin D supplementation to prevent acute respiratory infections: meta-analysis. BMJ. 2017;356:i6583. [8] Trexler ET, et al. ISSN position stand: beta-alanine. J Int Soc Sports Nutr. 2015;12:30.